



MAT1320-Linear Algebra

Lecture Notes

Change of Basis and Coordinate Transformations

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Coordinates

Theorem

Let V be vector space of dimension n and with basis

$\mathcal{B} = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$. Then any vector $\vec{w} \in V$ can be expressed uniquely as a linear combination of basis vectors in $\mathcal{B}$, say

$$\vec{w} = x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n.$$

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$$\vec{w} = x_1 \vec{v}_1 + x_2 \vec{v}_2 + \dots + x_n \vec{v}_n.$$

Note: These n scalars $x_1, x_2, \dots, x_n$ are called the **coordinates** of $\vec{w}$ relative to the basis $\mathcal{B}$, and they form a vector $(x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$ called the **coordinate vector** of $\vec{w}$ relative to $\mathcal{B}$. We denote this vector by $[\vec{w}]_{\mathcal{B}}$, or simply $[\vec{w}]$, when $\mathcal{B}$ is understood. Thus,

$$[\vec{w}]_{\mathcal{B}} = (x_1, x_2, \dots, x_n).$$

Example

Find coordinate vector of $\vec{\mathbf{w}} = (3, 2, 1) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

Example

Find coordinate vector of $\vec{w} = (3, 2, 1) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. For all $(x, y, z) \in \mathbb{R}^3$

$$\overset{3}{x} \overset{2}{y} \overset{1}{z} = \overset{3}{x_1}(1, 0, 0) + \overset{2}{x_2}(0, 1, 0) + \overset{1}{x_3}(0, 0, 1)$$

$$[\vec{w}]_{\mathcal{B}} = (3, 2, 1)$$

Example

Find coordinate vector of $\vec{\mathbf{w}} = (3, 2, 1) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$. For all $(x, y, z) \in \mathbb{R}^3$

$$(x, y, z) = x_1(1, 0, 0) + x_2(0, 1, 0) + x_3(0, 0, 1)$$

we have $x_1 = x$, $x_2 = y$, $x_3 = z$.

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we have $x_1 = x$, $x_2 = y$, $x_3 = z$. Then

$$(3, 2, 1) = 3.(1, 0, 0) + 2.(0, 1, 0) + 1.(0, 0, 1)$$

and so $[\vec{\mathbf{w}}]_{\mathcal{B}} = (3, 2, 1)$.

Coordinates

Example

Find coordinate vector of $\vec{w} = (1, 2, 3) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 2, 0), (2, 0, 1), (1, 2, 1)\}$.

$$a \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + c \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

$$a + 2b + c = 1$$

$$2a + 2c = 2$$

$$b + c = 3$$

$$a + c = 1$$

$$b + c = 3$$

$$a = 1 - c$$

$$b = 3 - c$$

$$1 - c + 6 - 2c = 1$$

$$7 - 2c = 1$$

$$6 = 2c$$

$$c = 3$$

$$b = 0$$

$$a = -2$$

Example

Find coordinate vector of $\vec{\mathbf{w}} = (1, 2, 3) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 2, 0), (2, 0, 1), (1, 2, 1)\}$. For all $(x, y, z) \in \mathbb{R}^3$

$$(x, y, z) = x_1(1, 2, 0) + x_2(2, 0, 1) + x_3(1, 2, 1)$$

Example

Find coordinate vector of $\vec{w} = (1, 2, 3) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 2, 0), (2, 0, 1), (1, 2, 1)\}$. For all $(x, y, z) \in \mathbb{R}^3$

$$(x, y, z) = x_1(1, 2, 0) + x_2(2, 0, 1) + x_3(1, 2, 1)$$

$$x_1 + 2x_2 + x_3 = x$$

$$2x_1 + 2x_3 = y$$

$$x_2 + x_3 = z$$

Example

Find coordinate vector of $\vec{w} = (1, 2, 3) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 2, 0), (2, 0, 1), (1, 2, 1)\}$. For all $(x, y, z) \in \mathbb{R}^3$

$$\begin{aligned}(x, y, z) &= x_1(1, 2, 0) + x_2(2, 0, 1) + x_3(1, 2, 1) \\ \begin{aligned} x_1 + 2x_2 + x_3 &= x \\ 2x_1 + 2x_3 &= y \\ x_2 + x_3 &= z \end{aligned} &\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & x \\ 2 & 0 & 2 & y \\ 0 & 1 & 1 & z \end{array} \right)\end{aligned}$$

Coordinates

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Find coordinate vector of $\vec{w} = (1, 2, 3) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 2, 0), (2, 0, 1), (1, 2, 1)\}$. For all $(x, y, z) \in \mathbb{R}^3$

$$\begin{aligned}(x, y, z) &= x_1(1, 2, 0) + x_2(2, 0, 1) + x_3(1, 2, 1) \\ \begin{aligned} x_1 + 2x_2 + x_3 &= x \\ 2x_1 + 2x_3 &= y \\ x_2 + x_3 &= z \end{aligned} &\Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & x \\ 2 & 0 & 2 & y \\ 0 & 1 & 1 & z \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{2x+y-4z}{4} \\ 0 & 1 & 0 & \frac{2x-y}{4} \\ 0 & 0 & 1 & \frac{-2x+y+4z}{4} \end{array} \right)\end{aligned}$$

Coordinates

Example

Find coordinate vector of $\vec{w} = (1, 2, 3) \in \mathbb{R}^3$ relative to basis $\mathcal{B} = \{(1, 2, 0), (2, 0, 1), (1, 2, 1)\}$. For all $(x, y, z) \in \mathbb{R}^3$

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we have $x_1 = \frac{2x+y-4z}{4}$, $x_2 = \frac{2x-y}{4}$, $x_3 = \frac{-2x+y+4z}{4}$.

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$$\begin{aligned}(x, y, z) &= x_1(1, 2, 0) + x_2(2, 0, 1) + x_3(1, 2, 1) \\ x_1 + 2x_2 + x_3 &= x \\ 2x_1 + 2x_3 &= y \\ x_2 + x_3 &= z\end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & x \\ 2 & 0 & 2 & y \\ 0 & 1 & 1 & z \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{2x+y-4z}{4} \\ 0 & 1 & 0 & \frac{2x-y}{4} \\ 0 & 0 & 1 & \frac{-2x+y+4z}{4} \end{array} \right)$$

we have $x_1 = \frac{2x+y-4z}{4}$, $x_2 = \frac{2x-y}{4}$, $x_3 = \frac{-2x+y+4z}{4}$. Thus

$$(1, 2, 3) = -2 \cdot (1, 2, 0) + 0 \cdot (2, 0, 1) + 3 \cdot (1, 2, 1)$$

and $[\vec{w}]_{\mathcal{B}} = (-2, 0, 3)$.

$$\begin{array}{r} -2 + 2 + 3 \\ \hline = 3 \end{array}$$



Change of Basis and Coordinate Transformations Matrix

$$[w]_{B_2} = [M]_{B_1}^{B_2} \cdot [w]_{B_1}$$

$$\checkmark [w]_{B_1} = [M]_{B_2}^{B_1} \cdot [w]_{B_2}$$

Change of Basis and Coordinate Transformations

Definition

Let V be an n -space with two ordered basis

$$\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \text{ and } \mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}.$$

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$$\vec{v}_1 = a_{11}\vec{w}_1 + a_{21}\vec{w}_2 + \dots + a_{n1}\vec{w}_n$$

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$$\vec{v}_n = a_{1n}\vec{w}_1 + a_{2n}\vec{w}_2 + \dots + a_{nn}\vec{w}_n$$

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$$\begin{aligned}\vec{v}_1 &= a_{11}\vec{w}_1 + a_{21}\vec{w}_2 + \dots + a_{n1}\vec{w}_n \\ \vec{v}_2 &= a_{12}\vec{w}_1 + a_{22}\vec{w}_2 + \dots + a_{n2}\vec{w}_n \\ &\vdots \\ \vec{v}_n &= a_{1n}\vec{w}_1 + a_{2n}\vec{w}_2 + \dots + a_{nn}\vec{w}_n\end{aligned}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

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The matrix A is called **coordinates transformation matrix** from

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basis $\mathcal{B}_1$ to basis $\mathcal{B}_2$ and denoted by $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2}$.

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$$[M]_{\mathcal{B}_2}^{\mathcal{B}_1}$$

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$$\vec{w}_1 = b_{11} \vec{v}_1 + b_{21} \vec{v}_2 + \dots + b_{n1} \vec{v}_n$$

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$$\vdots$$

$$\vec{w}_n = b_{1n}\vec{v}_1 + b_{2n}\vec{v}_2 + \dots + b_{nn}\vec{v}_n$$

$$B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nn} \end{pmatrix} = [M]_{\mathcal{B}_2}^{\mathcal{B}_1}$$

Change of Basis and Coordinate Transformations

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$$B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nn} \end{pmatrix}$$

The matrix B is called **coordinates transformation matrix** from

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basis $\mathcal{B}_2$ to basis $\mathcal{B}_1$ and denoted by $B = [M]_{\mathcal{B}_2}^{\mathcal{B}_1}$.

Change of Basis and Coordinate Transformations

Theorem

Let V be an n -space with two ordered basis

$$\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \text{ and } \mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}.$$

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$\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ and $\mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \dots, \vec{w}_n\}$. Also, let

$A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2}$ and $B = [M]_{\mathcal{B}_2}^{\mathcal{B}_1}$ be the coordinates transformation

matrices from basis $\mathcal{B}_1$ to $\mathcal{B}_2$ and from basis $\mathcal{B}_2$ to $\mathcal{B}_1$ respectively.

Change of Basis and Coordinate Transformations

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matrices from basis $\mathcal{B}_1$ to $\mathcal{B}_2$ and from basis $\mathcal{B}_2$ to $\mathcal{B}_1$ respectively.

Then we have the following assertions:

1. $AB = I_n$ or $B = A^{-1}$ i.e., $\left([M]_{\mathcal{B}_1}^{\mathcal{B}_2}\right) \left([M]_{\mathcal{B}_2}^{\mathcal{B}_1}\right) = I_n$ or $\left([M]_{\mathcal{B}_2}^{\mathcal{B}_1}\right) = \left([M]_{\mathcal{B}_1}^{\mathcal{B}_2}\right)^{-1}$,

Change of Basis and Coordinate Transformations

$$[\vec{u}]_{\mathcal{B}_2} = \checkmark \quad [M]_{\mathcal{B}_1}^{\mathcal{B}_2} \checkmark \quad \rightarrow \underline{[A]_{\mathcal{B}_1} ?}$$

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Then we have the following assertions:

✓ 1. $AB = I_n$ or $B = A^{-1}$ i.e., $\left([M]_{\mathcal{B}_1}^{\mathcal{B}_2}\right) \left([M]_{\mathcal{B}_2}^{\mathcal{B}_1}\right) = I_n$ or

$$\left([M]_{\mathcal{B}_2}^{\mathcal{B}_1}\right) = \left([M]_{\mathcal{B}_1}^{\mathcal{B}_2}\right)^{-1},$$

✓ 2. For all $\vec{u}, \vec{v} \in V$ we have $\underbrace{[\vec{u}]_{\mathcal{B}_1}}_{\text{circled}} = \underbrace{[M]_{\mathcal{B}_2}^{\mathcal{B}_1}}_{\text{circled}} \underbrace{[\vec{u}]_{\mathcal{B}_2}}_{\text{circled}} \text{ and }$
$$\underbrace{[\vec{v}]_{\mathcal{B}_2}}_{\text{circled}} = \underbrace{[M]_{\mathcal{B}_1}^{\mathcal{B}_2}}_{\text{circled}} \underbrace{[\vec{v}]_{\mathcal{B}_1}}_{\text{circled}}.$$

$$[M]_{\mathcal{B}_2}^{\mathcal{B}_1} = \left([M]_{\mathcal{B}_1}^{\mathcal{B}_2}\right)^{-1}$$

Change of Basis and Coordinate Transformations

Example

Let

$$\mathcal{B}_1 = \{ \vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (1, 2, 1) \},$$

$$\mathcal{B}_2 = \{ \vec{w}_1 = (0, 2, 1), \vec{w}_2 = (-1, 0, 1), \vec{w}_3 = (-1, 3, 0) \}$$

be two ordered bases of V and $\vec{u} = (2, 3, 5)$ be the coordinates with respect to standard basis. Find each of the following.

1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2}$,

$$[M]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{bmatrix} 1 & 9/5 & 11/5 \\ -1 & -4/5 & -3/5 \\ 0 & -6/5 & -2/5 \end{bmatrix}$$

$$\vec{v}_1 = 1 \cdot \vec{w}_1 - 1 \cdot \vec{w}_2 + 0 \cdot \vec{w}_3$$

$$\vec{v}_2 = \frac{9}{5} \cdot \vec{w}_1 - \frac{4}{5} \cdot \vec{w}_2 - \frac{3}{5} \cdot \vec{w}_3$$

$$\vec{v}_3 = \frac{11}{5} \cdot \vec{w}_1 - \frac{3}{5} \cdot \vec{w}_2 - \frac{2}{5} \cdot \vec{w}_3$$

Change of Basis and Coordinate Transformations

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be two ordered bases of V and $\vec{u} = (2, 3, 5)$ be the coordinates with respect to standard basis. Find each of the following.

1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2},$

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1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2},$

2. $B = [M]_{\mathcal{B}_2}^{\mathcal{B}_1},$

3. $[\vec{u}]_{\mathcal{B}_1},$

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be two ordered bases of V and $\vec{u} = (2, 3, 5)$ be the coordinates with respect to standard basis. Find each of the following.

1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2},$

2. $B = [M]_{\mathcal{B}_2}^{\mathcal{B}_1},$

3. $[\vec{u}]_{\mathcal{B}_1},$

4. $[\vec{u}]_{\mathcal{B}_2}.$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_1 = a_{11}\vec{\mathbf{w}}_1 + a_{21}\vec{\mathbf{w}}_2 + a_{31}\vec{\mathbf{w}}_3$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\begin{aligned}\vec{\mathbf{v}}_1 &= a_{11}\vec{\mathbf{w}}_1 + a_{21}\vec{\mathbf{w}}_2 + a_{31}\vec{\mathbf{w}}_3 \\ (1, 2, 0) &= a_{11}(0, 2, 1) + a_{21}(-1, 0, 1) + a_{31}(-1, 3, 0)\end{aligned}$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_1 = a_{11}\vec{\mathbf{w}}_1 + a_{21}\vec{\mathbf{w}}_2 + a_{31}\vec{\mathbf{w}}_3$$

$$(1, 2, 0) = a_{11}(0, 2, 1) + a_{21}(-1, 0, 1) + a_{31}(-1, 3, 0)$$

$$-a_{21} - a_{31} = 1$$

$$2a_{11} + 3a_{31} = 2$$

$$a_{11} + a_{21} = 0$$

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1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

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$$(1, 2, 0) = a_{11}(0, 2, 1) + a_{21}(-1, 0, 1) + a_{31}(-1, 3, 0)$$

$$\begin{aligned} -a_{21} - a_{31} &= 1 \\ 2a_{11} + 3a_{31} &= 2 \\ a_{11} + a_{21} &= 0 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 1 \\ 2 & 0 & 3 & 2 \\ 1 & 1 & 0 & 0 \end{array} \right)$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

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$$(1, 2, 0) = a_{11}(0, 2, 1) + a_{21}(-1, 0, 1) + a_{31}(-1, 3, 0)$$

$$\begin{aligned} -a_{21} - a_{31} &= 1 \\ 2a_{11} + 3a_{31} &= 2 \\ a_{11} + a_{21} &= 0 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 1 \\ 2 & 0 & 3 & 2 \\ 1 & 1 & 0 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_1 = a_{11}\vec{\mathbf{w}}_1 + a_{21}\vec{\mathbf{w}}_2 + a_{31}\vec{\mathbf{w}}_3$$

$$(1, 2, 0) = a_{11}(0, 2, 1) + a_{21}(-1, 0, 1) + a_{31}(-1, 3, 0)$$

$$\begin{array}{rcl} -a_{21} - a_{31} = 1 \\ 2a_{11} + 3a_{31} = 2 \\ a_{11} + a_{21} = 0 \end{array} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 1 \\ 2 & 0 & 3 & 2 \\ 1 & 1 & 0 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

$$a_{11} = 1, a_{21} = -1, a_{31} = 0$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{v}_2 = a_{12}\vec{w}_1 + a_{22}\vec{w}_2 + a_{32}\vec{w}_3$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\begin{aligned}\vec{\mathbf{v}}_2 &= a_{12}\vec{\mathbf{w}}_1 + a_{22}\vec{\mathbf{w}}_2 + a_{32}\vec{\mathbf{w}}_3 \\ (2, 0, 1) &= a_{12}(0, 2, 1) + a_{22}(-1, 0, 1) + a_{32}(-1, 3, 0)\end{aligned}$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_2 = a_{12}\vec{\mathbf{w}}_1 + a_{22}\vec{\mathbf{w}}_2 + a_{32}\vec{\mathbf{w}}_3$$

$$(2, 0, 1) = a_{12}(0, 2, 1) + a_{22}(-1, 0, 1) + a_{32}(-1, 3, 0)$$

$$-a_{22} - a_{32} = 2$$

$$2a_{12} + 3a_{32} = 0$$

$$a_{12} + a_{22} = 1$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_2 = a_{12}\vec{\mathbf{w}}_1 + a_{22}\vec{\mathbf{w}}_2 + a_{32}\vec{\mathbf{w}}_3$$

$$(2, 0, 1) = a_{12}(0, 2, 1) + a_{22}(-1, 0, 1) + a_{32}(-1, 3, 0)$$

$$\begin{aligned} -a_{22} - a_{32} &= 2 \\ 2a_{12} + 3a_{32} &= 0 \\ a_{12} + a_{22} &= 1 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 2 \\ 2 & 0 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right)$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_2 = a_{12}\vec{\mathbf{w}}_1 + a_{22}\vec{\mathbf{w}}_2 + a_{32}\vec{\mathbf{w}}_3$$

$$(2, 0, 1) = a_{12}(0, 2, 1) + a_{22}(-1, 0, 1) + a_{32}(-1, 3, 0)$$

$$\begin{aligned} -a_{22} - a_{32} &= 2 \\ 2a_{12} + 3a_{32} &= 0 \\ a_{12} + a_{22} &= 1 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 2 \\ 2 & 0 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{9}{5} \\ 0 & 1 & 0 & -\frac{4}{5} \\ 0 & 0 & 1 & -\frac{6}{5} \end{array} \right)$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_2 = a_{12}\vec{\mathbf{w}}_1 + a_{22}\vec{\mathbf{w}}_2 + a_{32}\vec{\mathbf{w}}_3$$

$$(2, 0, 1) = a_{12}(0, 2, 1) + a_{22}(-1, 0, 1) + a_{32}(-1, 3, 0)$$

$$\begin{aligned} -a_{22} - a_{32} &= 2 \\ 2a_{12} + 3a_{32} &= 0 \\ a_{12} + a_{22} &= 1 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 2 \\ 2 & 0 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{9}{5} \\ 0 & 1 & 0 & -\frac{4}{5} \\ 0 & 0 & 1 & -\frac{6}{5} \end{array} \right)$$

$$a_{12} = \frac{9}{5}, a_{22} = -\frac{4}{5}, a_{32} = -\frac{6}{5}$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_3 = a_{13}\vec{\mathbf{w}}_1 + a_{23}\vec{\mathbf{w}}_2 + a_{33}\vec{\mathbf{w}}_3$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\begin{aligned}\vec{\mathbf{v}}_3 &= a_{13}\vec{\mathbf{w}}_1 + a_{23}\vec{\mathbf{w}}_2 + a_{33}\vec{\mathbf{w}}_3 \\ (1, 2, 1) &= a_{13}(0, 2, 1) + a_{23}(-1, 0, 1) + a_{33}(-1, 3, 0)\end{aligned}$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_3 = a_{13}\vec{\mathbf{w}}_1 + a_{23}\vec{\mathbf{w}}_2 + a_{33}\vec{\mathbf{w}}_3$$

$$(1, 2, 1) = a_{13}(0, 2, 1) + a_{23}(-1, 0, 1) + a_{33}(-1, 3, 0)$$

$$-a_{23} - a_{33} = 1$$

$$2a_{13} + 3a_{33} = 2$$

$$a_{13} + a_{23} = 1$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_3 = a_{13}\vec{\mathbf{w}}_1 + a_{23}\vec{\mathbf{w}}_2 + a_{33}\vec{\mathbf{w}}_3$$

$$(1, 2, 1) = a_{13}(0, 2, 1) + a_{23}(-1, 0, 1) + a_{33}(-1, 3, 0)$$

$$\begin{aligned} -a_{23} - a_{33} &= 1 \\ 2a_{13} + 3a_{33} &= 2 \\ a_{13} + a_{23} &= 1 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 1 \\ 2 & 0 & 3 & 2 \\ 1 & 1 & 0 & 1 \end{array} \right)$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_3 = a_{13}\vec{\mathbf{w}}_1 + a_{23}\vec{\mathbf{w}}_2 + a_{33}\vec{\mathbf{w}}_3$$

$$(1, 2, 1) = a_{13}(0, 2, 1) + a_{23}(-1, 0, 1) + a_{33}(-1, 3, 0)$$

$$\begin{aligned} -a_{23} - a_{33} &= 1 \\ 2a_{13} + 3a_{33} &= 2 \\ a_{13} + a_{23} &= 1 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 1 \\ 2 & 0 & 3 & 2 \\ 1 & 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{8}{5} \\ 0 & 1 & 0 & -\frac{3}{5} \\ 0 & 0 & 1 & -\frac{2}{5} \end{array} \right)$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$\vec{\mathbf{v}}_3 = a_{13}\vec{\mathbf{w}}_1 + a_{23}\vec{\mathbf{w}}_2 + a_{33}\vec{\mathbf{w}}_3$$

$$(1, 2, 1) = a_{13}(0, 2, 1) + a_{23}(-1, 0, 1) + a_{33}(-1, 3, 0)$$

$$\begin{aligned} -a_{23} - a_{33} &= 1 \\ 2a_{13} + 3a_{33} &= 2 \\ a_{13} + a_{23} &= 1 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 1 \\ 2 & 0 & 3 & 2 \\ 1 & 1 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{8}{5} \\ 0 & 1 & 0 & -\frac{3}{5} \\ 0 & 0 & 1 & -\frac{2}{5} \end{array} \right)$$
$$a_{13} = \frac{8}{5}, a_{23} = -\frac{3}{5}, a_{33} = -\frac{2}{5}$$

Change of Basis and Coordinate Transformations

1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

Change of Basis and Coordinate Transformations

1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & \frac{9}{5} & \frac{8}{5} \\ -1 & -\frac{4}{5} & -\frac{3}{5} \\ 0 & -\frac{6}{5} & -\frac{2}{5} \end{pmatrix}$$

Change of Basis and Coordinate Transformations

1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & \frac{9}{5} & \frac{8}{5} \\ -1 & -\frac{4}{5} & -\frac{3}{5} \\ 0 & -\frac{6}{5} & -\frac{2}{5} \end{pmatrix}$$

2. $B = [M]_{\mathcal{B}_2}^{\mathcal{B}_1}$:

$$B = ([M]_{\mathcal{B}_2}^{\mathcal{B}_1}) = ([M]_{\mathcal{B}_1}^{\mathcal{B}_2})^{-1} = A^{-1}$$

Change of Basis and Coordinate Transformations

1. $A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$A = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1 & \frac{9}{5} & \frac{8}{5} \\ -1 & -\frac{4}{5} & -\frac{3}{5} \\ 0 & -\frac{6}{5} & -\frac{2}{5} \end{pmatrix}$$

2. $B = [M]_{\mathcal{B}_2}^{\mathcal{B}_1}$:

$$\begin{aligned} B &= \left([M]_{\mathcal{B}_2}^{\mathcal{B}_1} \right) = \left([M]_{\mathcal{B}_1}^{\mathcal{B}_2} \right)^{-1} = A^{-1} \\ &= \begin{pmatrix} -\frac{1}{2} & -\frac{3}{2} & \frac{1}{4} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{5}{4} \\ \frac{3}{2} & \frac{3}{2} & \frac{5}{4} \end{pmatrix}. \end{aligned}$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{\mathcal{B}_1}$: $\mathcal{B}_1 = \{\vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (1, 2, 1)\}$

$$\vec{u} = y_1 \vec{v}_1 + y_2 \vec{v}_2 + y_3 \vec{v}_3$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{\mathcal{B}_1}$: $\mathcal{B}_1 = \{ \vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (1, 2, 1) \}$

$$\vec{u} = y_1 \vec{v}_1 + y_2 \vec{v}_2 + y_3 \vec{v}_3$$

$$(2, 3, 5) = y_1 (1, 2, 0) + y_2 (2, 0, 1) + y_3 (1, 2, 1)$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{\mathcal{B}_1}$: $\mathcal{B}_1 = \{\vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (1, 2, 1)\}$

$$\vec{u} = y_1 \vec{v}_1 + y_2 \vec{v}_2 + y_3 \vec{v}_3$$

$$(2, 3, 5) = y_1 (1, 2, 0) + y_2 (2, 0, 1) + y_3 (1, 2, 1)$$

$$y_1 + 2y_2 + y_3 = 2$$

$$2y_1 + 2y_3 = 3$$

$$y_2 + y_3 = 5$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{\mathcal{B}_1}$: $\mathcal{B}_1 = \{\vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (1, 2, 1)\}$

$$\vec{u} = y_1 \vec{v}_1 + y_2 \vec{v}_2 + y_3 \vec{v}_3$$

$$(2, 3, 5) = y_1 (1, 2, 0) + y_2 (2, 0, 1) + y_3 (1, 2, 1)$$

$$\begin{aligned} y_1 + 2y_2 + y_3 &= 2 \\ 2y_1 + 2y_3 &= 3 \\ y_2 + y_3 &= 5 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 0 & 2 & 3 \\ 0 & 1 & 1 & 5 \end{array} \right)$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{B_1}$: $B_1 = \{\vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (1, 2, 1)\}$

$$\vec{u} = y_1 \vec{v}_1 + y_2 \vec{v}_2 + y_3 \vec{v}_3$$

$$(2, 3, 5) = y_1 (1, 2, 0) + y_2 (2, 0, 1) + y_3 (1, 2, 1)$$

$$\begin{aligned} y_1 + 2y_2 + y_3 &= 2 \\ 2y_1 + 2y_3 &= 3 \\ y_2 + y_3 &= 5 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 0 & 2 & 3 \\ 0 & 1 & 1 & 5 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{13}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{19}{4} \end{array} \right) \begin{matrix} y_1 \\ y_2 \\ y_3 \end{matrix}$$

$$[\vec{u}]_{B_1} = \left(-\frac{13}{4}, \frac{1}{4}, \frac{19}{4} \right)$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{B_1}$: $B_1 = \{\vec{v}_1 = (1, 2, 0), \vec{v}_2 = (2, 0, 1), \vec{v}_3 = (1, 2, 1)\}$

$$\vec{u} = y_1 \vec{v}_1 + y_2 \vec{v}_2 + y_3 \vec{v}_3$$

$$(2, 3, 5) = y_1 (1, 2, 0) + y_2 (2, 0, 1) + y_3 (1, 2, 1)$$

$$\begin{aligned} y_1 + 2y_2 + y_3 &= 2 \\ 2y_1 + 2y_3 &= 3 \\ y_2 + y_3 &= 5 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 0 & 2 & 3 \\ 0 & 1 & 1 & 5 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{13}{4} \\ 0 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{19}{4} \end{array} \right)$$

$$y_1 = -\frac{13}{4}, y_2 = \frac{1}{4}, y_3 = \frac{19}{4} \Rightarrow [\vec{u}]_{B_1} = \left(-\frac{13}{4}, \frac{1}{4}, \frac{19}{4} \right).$$

Change of Basis and Coordinate Transformations

4. $[\vec{u}]_{\mathcal{B}_2}$:

$$\mathcal{B}_2 = \{\vec{w}_1 = (0, 2, 1), \vec{w}_2 = (-1, 0, 1), \vec{w}_3 = (-1, 3, 0)\}$$

$$\vec{u} = x_1 \vec{w}_1 + x_2 \vec{w}_2 + x_3 \vec{w}_3$$

$$[\vec{u}]_{\mathcal{B}_2} = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} \cdot [\vec{u}]_{\mathcal{B}_1}$$

Change of Basis and Coordinate Transformations

4. $[\vec{u}]_{\mathcal{B}_2}$:

$$\mathcal{B}_2 = \{\vec{w}_1 = (0, 2, 1), \vec{w}_2 = (-1, 0, 1), \vec{w}_3 = (-1, 3, 0)\}$$

$$\vec{u} = x_1 \vec{w}_1 + x_2 \vec{w}_2 + x_3 \vec{w}_3$$

$$(2, 3, 5) = x_1 (0, 2, 1) + x_2 (-1, 0, 1) + x_3 (-1, 3, 0)$$

Change of Basis and Coordinate Transformations

4. $[\vec{u}]_{\mathcal{B}_2}$:

$$\mathcal{B}_2 = \{\vec{w}_1 = (0, 2, 1), \vec{w}_2 = (-1, 0, 1), \vec{w}_3 = (-1, 3, 0)\}$$

$$\vec{u} = x_1 \vec{w}_1 + x_2 \vec{w}_2 + x_3 \vec{w}_3$$

$$(2, 3, 5) = x_1 (0, 2, 1) + x_2 (-1, 0, 1) + x_3 (-1, 3, 0)$$

$$-x_2 - x_3 = 1$$

$$2x_1 + 3x_3 = 3$$

$$x_1 + x_2 = 5$$

Change of Basis and Coordinate Transformations

4. $[\vec{u}]_{\mathcal{B}_2}$:

$$\mathcal{B}_2 = \{\vec{w}_1 = (0, 2, 1), \vec{w}_2 = (-1, 0, 1), \vec{w}_3 = (-1, 3, 0)\}$$

$$\vec{u} = x_1 \vec{w}_1 + x_2 \vec{w}_2 + x_3 \vec{w}_3$$

$$(2, 3, 5) = x_1 (0, 2, 1) + x_2 (-1, 0, 1) + x_3 (-1, 3, 0)$$

$$\begin{array}{rcl} -x_2 - x_3 & = & 1 \\ 2x_1 + 3x_3 & = & 3 \\ x_1 + x_2 & = & 5 \end{array} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 2 \\ 2 & 0 & 3 & 3 \\ 1 & 1 & 0 & 5 \end{array} \right)$$

Change of Basis and Coordinate Transformations

4. $[\vec{u}]_{\mathcal{B}_2}$:

$$\mathcal{B}_2 = \{\vec{w}_1 = (0, 2, 1), \vec{w}_2 = (-1, 0, 1), \vec{w}_3 = (-1, 3, 0)\}$$

$$\vec{u} = x_1 \vec{w}_1 + x_2 \vec{w}_2 + x_3 \vec{w}_3$$

$$(2, 3, 5) = x_1 (0, 2, 1) + x_2 (-1, 0, 1) + x_3 (-1, 3, 0)$$

$$\begin{array}{rcl} -x_2 - x_3 = 1 \\ 2x_1 + 3x_3 = 3 \\ x_1 + x_2 = 5 \end{array} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 2 \\ 2 & 0 & 3 & 3 \\ 1 & 1 & 0 & 5 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{24}{5} \\ 0 & 1 & 0 & \frac{1}{5} \\ 0 & 0 & 1 & -\frac{11}{5} \end{array} \right)$$

Change of Basis and Coordinate Transformations

4. $[\vec{u}]_{\mathcal{B}_2}$:

$$\mathcal{B}_2 = \{\vec{w}_1 = (0, 2, 1), \vec{w}_2 = (-1, 0, 1), \vec{w}_3 = (-1, 3, 0)\}$$

$$\vec{u} = x_1 \vec{w}_1 + x_2 \vec{w}_2 + x_3 \vec{w}_3$$

$$(2, 3, 5) = x_1 (0, 2, 1) + x_2 (-1, 0, 1) + x_3 (-1, 3, 0)$$

$$\begin{aligned} -x_2 - x_3 &= 1 \\ 2x_1 + 3x_3 &= 3 \\ x_1 + x_2 &= 5 \end{aligned} \Rightarrow \left(\begin{array}{ccc|c} 0 & -1 & -1 & 2 \\ 2 & 0 & 3 & 3 \\ 1 & 1 & 0 & 5 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{24}{5} \\ 0 & 1 & 0 & \frac{1}{5} \\ 0 & 0 & 1 & -\frac{11}{5} \end{array} \right)$$

$$x_1 = \frac{24}{5}, x_2 = \frac{1}{5}, x_3 = -\frac{11}{5} \Rightarrow [\vec{u}]_{\mathcal{B}_2} = \left(\frac{24}{5}, \frac{1}{5}, -\frac{11}{5} \right).$$

Change of Basis and Coordinate Transformations

Notice that

$$[\vec{u}]_{\mathcal{B}_1} = [M]_{\mathcal{B}_2}^{\mathcal{B}_1} [\vec{u}]_{\mathcal{B}_2}$$

Change of Basis and Coordinate Transformations

Notice that

$$[\vec{u}]_{\mathcal{B}_1} = [M]_{\mathcal{B}_2}^{\mathcal{B}_1} [\vec{u}]_{\mathcal{B}_2} = \begin{pmatrix} -\frac{1}{2} & -\frac{3}{2} & \frac{1}{4} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{5}{4} \\ \frac{3}{2} & \frac{3}{2} & \frac{5}{4} \end{pmatrix} \begin{pmatrix} \frac{24}{5} \\ \frac{1}{5} \\ -\frac{11}{5} \end{pmatrix} = \begin{pmatrix} -\frac{13}{4} \\ \frac{1}{4} \\ \frac{19}{4} \end{pmatrix}$$

Change of Basis and Coordinate Transformations

Notice that

$$[\vec{u}]_{\mathcal{B}_1} = [M]_{\mathcal{B}_2}^{\mathcal{B}_1} [\vec{u}]_{\mathcal{B}_2} = \begin{pmatrix} -\frac{1}{2} & -\frac{3}{2} & \frac{1}{4} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{5}{4} \\ \frac{3}{2} & \frac{3}{2} & \frac{5}{4} \end{pmatrix} \begin{pmatrix} \frac{24}{5} \\ \frac{1}{5} \\ -\frac{11}{5} \end{pmatrix} = \begin{pmatrix} -\frac{13}{4} \\ \frac{1}{4} \\ \frac{19}{4} \end{pmatrix}$$

$$[\vec{u}]_{\mathcal{B}_2} = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} [\vec{u}]_{\mathcal{B}_1}$$

Change of Basis and Coordinate Transformations

Ex: B_1, B_2 are two bases for $\mathbb{R}^2$, not $\mathbb{R}^3$.

$$[\vec{u}]_{B_1} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad [\vec{u}]_{B_2} = \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix}^{-1} [\vec{u}]_{B_1} = [\vec{u}]_{B_2}$$

$$[\vec{u}]_{B_2} = [M]_{B_2}^{B_1} [\vec{u}]_{B_1}$$

$$[M]_{B_2}^{B_1} = ([M]_{B_1}^{B_2})^{-1} = \left(\begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix} \right)^{-1}$$

= ✓

Notice that

$$[\vec{u}]_{B_1} = [M]_{B_2}^{B_1} [\vec{u}]_{B_2} = \begin{pmatrix} -\frac{1}{2} & -\frac{3}{2} & \frac{1}{4} \\ -\frac{1}{2} & -\frac{1}{2} & -\frac{5}{4} \\ \frac{3}{2} & \frac{3}{2} & \frac{5}{4} \end{pmatrix} \begin{pmatrix} \frac{24}{5} \\ \frac{1}{5} \\ -\frac{11}{5} \end{pmatrix} = \begin{pmatrix} -\frac{13}{4} \\ \frac{1}{4} \\ \frac{19}{4} \end{pmatrix}$$

$$[\vec{u}]_{B_2} = [M]_{B_1}^{B_2} [\vec{u}]_{B_1} = \begin{pmatrix} 1 & \frac{9}{5} & \frac{8}{5} \\ -1 & -\frac{4}{5} & -\frac{3}{5} \\ 0 & -\frac{6}{5} & -\frac{2}{5} \end{pmatrix} \begin{pmatrix} -\frac{13}{4} \\ \frac{1}{4} \\ \frac{19}{4} \end{pmatrix} = \begin{pmatrix} \frac{24}{5} \\ \frac{1}{5} \\ -\frac{11}{5} \end{pmatrix}$$

Change of Basis and Coordinate Transformations

Example

Let $\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and $\mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ be two ordered bases of V such that

$$\vec{v}_1 = \vec{w}_1 + \vec{w}_2 - \vec{w}_3$$

Change of Basis and Coordinate Transformations

Example

Let $\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and $\mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ be two ordered bases of V such that

$$\begin{aligned}\vec{v}_1 &= \vec{w}_1 + \vec{w}_2 - \vec{w}_3 \\ \vec{v}_2 &= 2\vec{w}_1 - \vec{w}_2 + \vec{w}_3\end{aligned}$$

Change of Basis and Coordinate Transformations

Example

Let $\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and $\mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ be two ordered bases of V such that

$$\vec{v}_1 = \vec{w}_1 + \vec{w}_2 - \vec{w}_3$$

$$\vec{v}_2 = 2\vec{w}_1 - \vec{w}_2 + \vec{w}_3$$

$$\vec{v}_3 = \vec{w}_1 + 2\vec{w}_3.$$

Change of Basis and Coordinate Transformations

Example

Let $\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and $\mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ be two ordered bases of V such that

$$\vec{v}_1 = 1\vec{w}_1 + \vec{w}_2 - \vec{w}_3 \quad \checkmark$$

$$\vec{v}_2 = 2\vec{w}_1 - \vec{w}_2 + \vec{w}_3 \quad \checkmark$$

$$\vec{v}_3 = 1\vec{w}_1 + 2\vec{w}_3. \quad \checkmark$$

1. $[M]_{\mathcal{B}_1}^{\mathcal{B}_2} = ?$ $\begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 2 \end{bmatrix}$

2. $[M]_{\mathcal{B}_2}^{\mathcal{B}_1} = ([M]_{\mathcal{B}_1}^{\mathcal{B}_2})^{-1}$

Change of Basis and Coordinate Transformations

Example

Let $\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and $\mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ be two ordered bases of V such that

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Change of Basis and Coordinate Transformations

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1. $[M]_{\mathcal{B}_1}^{\mathcal{B}_2} = ?$

2. $[M]_{\mathcal{B}_2}^{\mathcal{B}_1} = ?$

3. $[\vec{u}]_{\mathcal{B}_1} = (1, 0, 3) \Rightarrow [\vec{u}]_{\mathcal{B}_2} = ?$

Change of Basis and Coordinate Transformations

Example

Let $\mathcal{B}_1 = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and $\mathcal{B}_2 = \{\vec{w}_1, \vec{w}_2, \vec{w}_3\}$ be two ordered bases of V such that

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1. $[M]_{\mathcal{B}_1}^{\mathcal{B}_2} = ?$

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3. $[\vec{u}]_{\mathcal{B}_1} = (1, 0, 3) \Rightarrow [\vec{u}]_{\mathcal{B}_2} = ?$

4. $[\vec{v}]_{\mathcal{B}_2} = (2, -1, -2) \Rightarrow [\vec{v}]_{\mathcal{B}_1} = ?$

$$[v,] = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$[\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 2 \end{pmatrix}$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$[\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 2 \end{pmatrix}$$

2. $\mathbf{B} = [\mathbf{M}]_{\mathcal{B}_2}^{\mathcal{B}_1}$:

$$\left([\mathbf{M}]_{\mathcal{B}_2}^{\mathcal{B}_1}\right) = \left([\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}\right)^{-1}$$

Change of Basis and Coordinate Transformations

1. $\mathbf{A} = [\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}$:

$$[\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 2 \end{pmatrix}$$

2. $\mathbf{B} = [\mathbf{M}]_{\mathcal{B}_2}^{\mathcal{B}_1}$:

$$\begin{aligned} \left([\mathbf{M}]_{\mathcal{B}_2}^{\mathcal{B}_1}\right) &= \left([\mathbf{M}]_{\mathcal{B}_1}^{\mathcal{B}_2}\right)^{-1} \\ &= \begin{pmatrix} \frac{1}{3} & \frac{1}{2} & -\frac{1}{6} \\ \frac{1}{3} & -\frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \end{aligned}$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{B_2}$:

$$[\vec{u}]_{B_2} = [M]_{B_1}^{B_2} [\vec{u}]_{B_1} \quad \checkmark$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{\mathcal{B}_2}$:

$$[\vec{u}]_{\mathcal{B}_2} = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} [\vec{u}]_{\mathcal{B}_1} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix}$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{B_2}$:

$$[\vec{u}]_{B_2} = [M]_{B_1}^{B_2} [\vec{u}]_{B_1} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix}$$

4. $[\vec{v}]_{B_1}$:

$$[\vec{v}]_{B_1} = [M]_{B_2}^{B_1} [\vec{v}]_{B_2}$$

Change of Basis and Coordinate Transformations

3. $[\vec{u}]_{\mathcal{B}_2}$:

$$[\vec{u}]_{\mathcal{B}_2} = [M]_{\mathcal{B}_1}^{\mathcal{B}_2} [\vec{u}]_{\mathcal{B}_1} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 0 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix}$$

4. $[\vec{v}]_{\mathcal{B}_1}$:

$$[\vec{v}]_{\mathcal{B}_1} = [M]_{\mathcal{B}_2}^{\mathcal{B}_1} [\vec{v}]_{\mathcal{B}_2} = \begin{pmatrix} \frac{1}{3} & \frac{1}{2} & -\frac{1}{6} \\ \frac{1}{3} & -\frac{1}{2} & -\frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{3}{2} \\ -\frac{3}{2} \end{pmatrix}$$

?